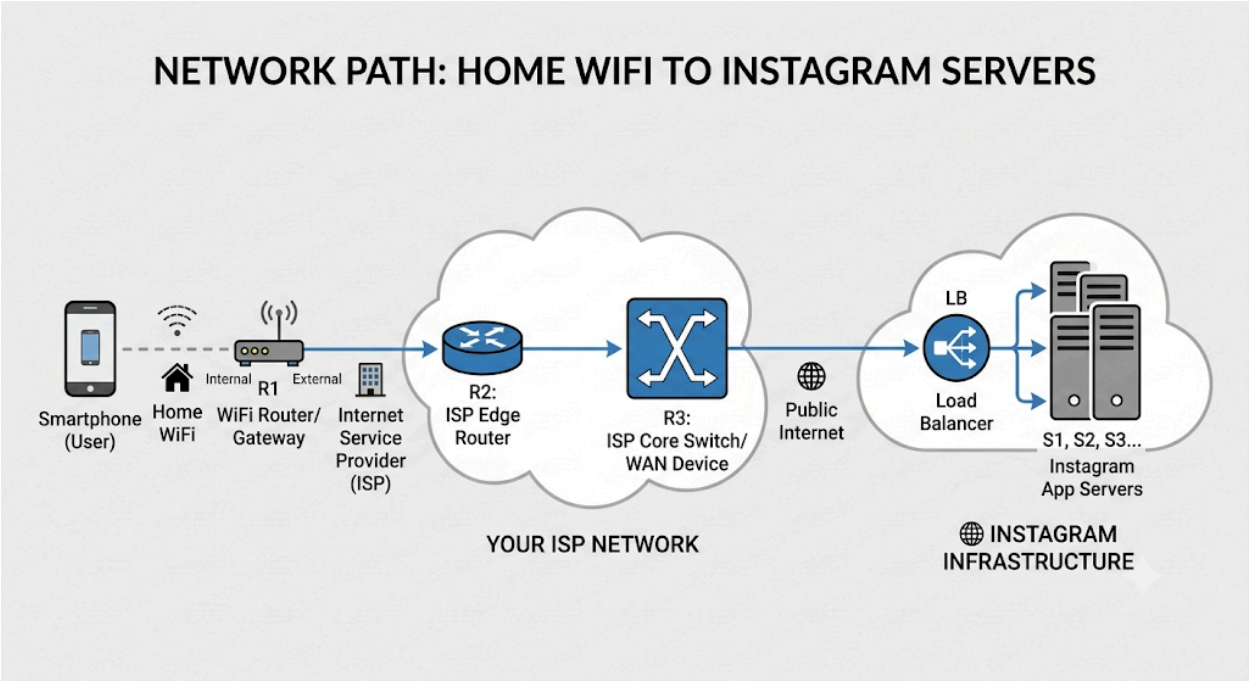
Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

ANS:



Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

ANS:

Here is a comparison table outlining the key differences in speed, cost, and real-world use cases for MPLS, Leased Line, and Broadband Internet:

WAN Technology	Speed & Performance	Cost	Real-Life Use Case
Broadband Internet	Variable speeds (often asymmetrical: faster download, slower upload); subject to	Low to moderate; economical for everyday users.	**Your Home** (streaming movies, casual browsing, online gaming).

	congestion during peak hours.		
Leased Line (ILL)	High, symmetrical speeds (equal upload and download) with guaranteed, unshared bandwidth and strict SLAs.	High; premium pricing due to dedicated physical infrastructure and guaranteed uptime.	**Zomato Corporate Office** (handling heavy merchant data syncs, video conferencing, and reliable cloud operations).
MPLS (Multi-Protocol Label Switching)	High-performance routing with minimal latency and built-in Quality of Service (QoS) traffic prioritization.	High (enterprise-grade); scales based on multi-site mesh routing complexity.	**IRCTC / Bank Regional Offices** (securely connecting multiple branch offices and data centers into a private internal network).

Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Hint: Focus on how the VPN creates a private tunnel over the public WAN.

ANS:

Encrypted Tunnel: Encrypts your app data on public Wi-Fi, turning it into unreadable code as it travels across the public WAN.

Threat Shield: Protects your traffic from packet-sniffing and man-in-the-middle attacks by local eavesdroppers.

IP Masking: Routes your traffic through a secure remote VPN server, hiding your real IP address from the destination service (like Spotify or YouTube).

Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

ANS:

For connecting large, high-throughput e-commerce hubs like Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi, **SD-WAN (Software-Defined Wide Area Network)** running over high-speed fiber or broadband links is the best choice.

Justification

1. **Cost-Effective Multi-Site Scalability & Failover:**
    - SD-WAN allows Flipkart to use a mix of cost-effective commercial internet links across these major cities instead of expensive, rigid dedicated circuits. It dynamically steers traffic and provides automatic backup routing if one city's link degrades, ensuring zero downtime for logistics operations.
2. **Centralized Management & Low Latency for Inventory Sync:**
    - E-commerce warehouses require real-time database updates for stock levels and order fulfillment. SD-WAN provides centralized software control to prioritize critical inventory traffic, ensuring fast, synchronized data flow between regional hubs like Mumbai, Delhi, and Ahmedabad.